Basic Ladder Safety on Construction Sites

OSHA-Compliant Ladder Safety Information

Applicable OSHA Standards

Ladder safety on construction sites is primarily regulated under:

- **29 CFR 1926 Subpart X — Stairways and Ladders**
- **29 CFR 1926.1053 — Ladders**
- **29 CFR 1926.1060 — Training Requirements**
- OSHA Portable Ladder Safety guidance

Ladders are among the most commonly used pieces of equipment on construction sites. Their frequent use can make them seem routine; however, improper ladder selection, setup, inspection, or use can result in serious falls and injuries.

Employers are responsible for ensuring that employees who use ladders receive appropriate training so they can recognize ladder-related hazards and understand the procedures necessary to minimize those hazards.

1. Common Ladder Hazards

Common ladder hazards include:

- Standing on the top or top step of a stepladder.
- Leaning excessively to one side.
- Using damaged or defective ladders.
- Placing ladders on uneven or unstable surfaces.
- Using ladders near energized electrical equipment or power lines.
- Carrying tools or materials while climbing that interfere with balance.
- Using a ladder that is too short for the task.
- Positioning ladders in doorways, walkways, or traffic areas without protection.
- Using a closed stepladder as a straight ladder when it was not designed for that purpose.
- Exceeding the manufacturer's load rating.
- Moving or adjusting a ladder while it is occupied.

Proper planning before ladder use can eliminate many of these hazards.

2. Selecting the Correct Ladder

Before using a ladder, select one that is appropriate for the height, working conditions, load, and type of work being performed.

Ladder Height

Use a ladder that allows the work to be completed without standing on prohibited steps or excessively reaching to the side.

Never place a ladder on another object to gain additional height, including:

- Boxes
- Buckets
- Lumber
- Tables
- Scaffolding
- Other unstable surfaces

If the ladder is not tall enough to complete the work safely, obtain the correct ladder.

Duty Rating and Load Capacity

Every ladder has a maximum intended load or manufacturer's rated capacity.

The total load includes more than the worker's body weight. Consider:

Worker + clothing + tools + equipment + materials supported by the ladder

Never load a ladder beyond its maximum intended load or manufacturer's rated capacity.

Ladder Type

Use a ladder only for the purpose for which it was designed.

A self-supporting stepladder should not be converted into or used as another type of ladder unless the manufacturer specifically designed it for that configuration.

3. Ladder Inspection

Ladders should be inspected before use.

Employees should examine:

- Side rails
- Steps and rungs
- Ladder feet
- Spreaders and locking mechanisms
- Extension locks
- Rope and pulley systems, when equipped
- Hardware and fasteners
- Labels and warning markings
- Fiberglass components
- Cracks, splits, bends, corrosion, or other structural damage
- Oil, grease, mud, water, or other slipping hazards

Ladders must remain free from oil, grease, and other slipping hazards.

OSHA also requires ladders to be periodically inspected by a competent person and after an occurrence that could affect their safe use.

Damaged or Defective Ladders

A ladder with structural damage or another condition that could affect safe use must not be used.

Defective portable ladders must be immediately identified as defective, such as by applying a **"DO NOT USE"** tag, and removed from service.

Repairs must restore the ladder to a condition meeting its original design criteria before it is returned to service.

When there is uncertainty regarding a ladder's condition:

Do not use the ladder.

Report the condition to a supervisor or designated competent person.

4. Proper Ladder Setup

A ladder that is in good condition can still become hazardous when it is set up incorrectly.

Stable and Level Surfaces

Ladders must be used on stable and level surfaces unless they are secured against accidental displacement.

Never use boxes, blocks, lumber, or other unstable objects beneath a ladder to increase its height or compensate for an uneven surface.

Stepladders

When using a stepladder:

- Fully open the ladder.
- Engage all spreaders and locking devices.
- Ensure all ladder feet are firmly supported.
- Do not climb a closed stepladder unless it was specifically designed for that configuration.
- Never use the top or top step as a step.

Stepladders must have a metal spreader or locking device that holds the front and back sections in an open position while the ladder is in use.

5. Extension and Straight Ladders — The 4-to-1 Rule

Non-self-supporting ladders should generally be positioned using approximately a **4-to-1 setup ratio**.

For every **4 feet of ladder working length**, the bottom of the ladder should be positioned approximately **1 foot away from the upper support**.

Example

For a ladder with approximately 20 feet of working length:

20 ÷ 4 = 5 feet

The base should therefore be positioned approximately 5 feet from the upper support.

This positioning helps provide a stable climbing angle.

6. Accessing Upper Levels — The 3-Foot Rule

When a portable ladder is used to access an upper landing, the ladder's side rails should generally extend at least:

3 FEET

above the upper landing surface.

If the ladder cannot extend at least 3 feet because of its length, the ladder must be secured at the top to a rigid support and a grasping device, such as a grabrail, must be provided to assist employees while getting on and off the ladder.

This requirement is particularly important when accessing:

- Roofs
- Platforms
- Mezzanines
- Elevated work areas

The transition between a ladder and an elevated surface is an area where workers can easily lose their balance.

7. Safe Climbing and Working Practices

Face the Ladder

Employees must face the ladder when ascending or descending.

Maintain Secure Contact

Employees must use at least one hand to grasp the ladder while climbing.

OSHA ladder safety guidance also recommends maintaining **three-point contact** while climbing whenever possible:

- Two hands and one foot or
- Two feet and one hand

Do Not Carry Loads That Affect Balance

Do not carry objects or loads that could cause loss of balance while climbing.

When materials or tools must be moved, consider:

- Tool belts
- Tool pouches
- Material lifts
- Hoisting equipment
- Having another employee safely pass or raise materials

Avoid Overreaching

Keep the body controlled and avoid excessive sideways reaching.

If the work cannot be comfortably reached:

Climb down and reposition the ladder.

Do not risk a fall simply to avoid moving the ladder.

Never Move an Occupied Ladder

Never move, shift, or extend a ladder while someone is standing or working on it.

8. Electrical Hazards

Electrical hazards require special attention whenever ladders are used around electrical systems, wiring, equipment, temporary power installations, or overhead power lines.

Before carrying, raising, or positioning a ladder, check the surrounding area for:

- Overhead electrical lines
- Exposed energized conductors
- Temporary electrical installations
- Open electrical equipment

- Energized equipment the ladder or employee could contact

Ladders must have **nonconductive side rails** when an employee or ladder could contact exposed energized electrical equipment, subject to applicable OSHA requirements and exceptions.

Metal ladders should not be used near power lines or exposed energized electrical equipment.

Important Jobsite Practice

Before raising or carrying a ladder, look up and look around.

9. Doorways, Walkways, and Traffic Areas

Additional precautions are required when ladders are positioned in areas where employees, vehicles, doors, or equipment could strike or displace them.

Use particular caution in:

- Doorways
- Hallways
- Walkways
- Driveways
- Loading areas
- Construction traffic areas

When workplace activities or traffic could displace a ladder, the ladder must either be:

- Secured against accidental displacement, or
- Protected by a barricade that keeps traffic and other activities away from the ladder

The areas around both the top and bottom of the ladder must remain clear.

Never assume someone on the opposite side of a closed door knows that a ladder is in use.

10. Essential Ladder Safety Rules

All employees using ladders should follow these fundamental rules:

1. Inspect the ladder before use.
2. Select the correct ladder for the task.
3. Know and follow the ladder's load rating.
4. Place ladders on stable and level surfaces.
5. Fully open and lock stepladders before use.
6. Never stand on the top or top step of a stepladder.
7. Use approximately a 4-to-1 setup ratio for straight and extension ladders.
8. Extend access ladders at least 3 feet above an upper landing when required.

9. Face the ladder when climbing.
10. Maintain secure contact and use three-point contact as a safe climbing practice.
11. Do not carry tools or materials that could cause loss of balance.
12. Do not overreach; climb down and reposition the ladder.
13. Keep ladders away from electrical hazards and use nonconductive side rails when required.
14. Secure or barricade ladders that are exposed to workplace traffic.
15. Tag damaged ladders **"DO NOT USE"** and remove them from service.
16. Never move or adjust a ladder while it is occupied.

11. When a Ladder Must Be Removed From Service

Immediately remove a ladder from service when it has a condition that could compromise its structural integrity or safe use.

Examples include:

- Cracked or split side rails
- Bent or damaged rungs
- Broken locking mechanisms
- Damaged feet
- Missing hardware
- Significant corrosion
- Damaged fiberglass
- Defective extension locks
- Damaged rope or pulley systems
- Other structural defects

The ladder should be clearly identified as defective and must not be returned to service until it has been properly repaired in accordance with its original design criteria.

Ladder Safety Summary

Most ladder incidents are preventable.

Many accidents result from decisions made before or during ladder use rather than from a sudden ladder failure. Common contributing factors include:

- Choosing the wrong ladder.
- Skipping the inspection.
- Setting the ladder up incorrectly.
- Overreaching.
- Carrying excessive or awkward loads.
- Working too close to electrical hazards.
- Continuing to use damaged equipment.